Profit = Revenue – (Vessel fixed cost + Vessel dynamic cost + Container load/unload costs + Transshipment costs + Port call costs + Port Escort fees)

Where **Revenue** = Container rate from origin to destination * Number of containers from origin to destination. This needs to be calculated for each leg in the service and each container type

Vessel fixed cost = Vessel insurance + Crew salary + Maintenance cost. Usually, liner companies calculate it per day fixed costs

Vessel dynamic cost – Vessel fuel consumption which is summation of idling fuel consumption and sailing fuel consumption. For each port, depending on historical sailing, we can find typically how much time it takes to dock from arrival + cargo operation times and use it as idling time (hours). For sailing fuel consumption, find out the distance between two ports and use sailing fuel consumption rate

Container load/unload costs – for each port, container load/unload cost vary. And even they vary by container type. Usually loading and loading costs are the same but some ports have different costs.

Transshipment costs – usually ports have the same costs as above load/unload costs for transshipment, but some ports charge additional fees if the containers are required to be stored before they are loaded onto next vessel. While some ports charge less fees. If this isn’t available in the dataset, we can assume +-20% as that of load/unload costs.

Port call costs – charged per vessel visit to the port and varies by port

Port Escort fees – these are required for tugs to bring the vessel from ocean to terminal after its main engine is shut off

In academic literature, we will find mostly Revenue, Vessel dynamic cost, Container load/unload costs and Transshipment costs.